Forecasting Electricity Demand and Renewable Generation: Why the Same Machine Learning Models Do Not Work for Both

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SUMMARY

Electricity grids need to know two things in advance: how much electricity people will use, and how much electricity will come from renewable sources. In our first project, we found that machine-learning models could improve 24-hour-ahead electricity demand forecasts compared with a simple previous-day estimate. In this project, we wanted to find out whether the same approach would work for renewable generation.

We used 2025 California Independent System Operator (CAISO) data for solar and wind generation and compared four approaches: a simple persistence method, Ridge Regression, Random Forest, and Gradient Boosting. We used the same test period and model settings as our demand study so that the comparison would be as fair as possible. We evaluated the models using mean absolute error (MAE), root mean squared error (RMSE), and mean absolute percentage error (MAPE).

The results were different from what we expected. For electricity demand, Gradient Boosting was the best model and reduced MAE by 31.7% compared with the persistence baseline. For renewable generation, however, Gradient Boosting and Random Forest both performed worse than the baseline. Ridge Regression was the only model that improved on the baseline, reducing MAE by 4.8%. This was an important result because the model that worked best for demand was not the best model for renewable generation.

Our results suggest that electricity demand and renewable generation have different patterns that affect how easy they are to predict. Demand is strongly influenced by daily routines, while renewable generation depends more on changing weather conditions. This means that a more complicated machine-learning model is not always better. More testing with separate solar and wind forecasts, different seasons, and data from other regions would help us understand whether these results continue to hold.

1. Introduction

A grid operator has to answer two separate questions about tomorrow. The first is how much electricity people are going to use. The second is how much of that electricity is going to show up on its own, from solar panels and wind turbines, without anyone deciding to generate it. The difference between those two numbers is what everything else has to cover — natural gas plants, hydro, imports, and storage.

In our first study we looked only at the first question. We compared five models for predicting electricity demand 24 hours ahead using 2025 data from the California Independent System Operator (CAISO), and gradient boosting worked best, cutting mean absolute error by 31.7% compared to a naive baseline [10]. That result made us curious about the second question. In our 2025 dataset, renewable generation accounted for 31.3% of total demand, so a demand forecast on its own only tells half the story.

So we asked: can we forecast renewable generation the same way? We already had solar and wind output in the same dataset, so we reused the exact same pipeline, the same test period, and the same model settings, and just changed what we were predicting.

Our research question for this paper is: do the same machine-learning models that worked for demand also work for renewable generation?

The short answer is no, and we think the reason why is the most interesting part of the paper.

2. Background

2.1 Why the Grid Needs Two Forecasts

Electricity supply and demand have to match at every moment, because there is no cheap way to store electricity at grid scale. Grid operators schedule power plants ahead of time based on what they expect demand to be. If they schedule too much, they waste fuel and money. If they schedule too little, they have to buy expensive emergency power, or in the worst case the grid becomes unstable.

Solar and wind change this problem. Their output is not scheduled by anyone — it depends on the weather. What operators actually need to plan around is net load, which is demand minus renewable generation. Net load is what conventional plants have to supply. Getting net load right means getting both forecasts right [2].

2.2 What Makes Demand Predictable

Electricity demand comes from people, and people are repetitive. Demand goes up in the morning when people wake up, dips in the middle of the day, peaks in the early evening, and drops overnight. Weekdays look different from weekends. Summer looks different from winter, mostly because of air conditioning.

Demand also has a kind of momentum. If demand is 25,000 MW right now, it is very unlikely to be 40,000 MW an hour from now. This is why a naive baseline that just repeats yesterday's value already does reasonably well, and it is also why a model with access to lag features and weather can do noticeably better.

2.3 What Makes Renewable Generation Harder to Predict

Renewable output works differently. Solar has one very strong pattern — it is zero at night, rises after sunrise, peaks around midday, and falls to zero again at sunset. But on top of that pattern, cloud cover can cut output sharply and clouds are not very predictable a day in advance. Wind has almost no daily pattern at all. It depends on wind speed, which changes for reasons that have nothing to do with the time of day.

Renewable output also has much less momentum than demand. Solar output right now tells you very little about solar output 24 hours from now, because tomorrow's clouds are independent of today's. We expected this to make forecasting harder. What we did not expect was that it would change which model won.

2.4 Models Used in This Study

We tested four approaches for renewable forecasting:

- Persistence baseline: predicts that renewable output 24 hours from now will equal renewable output at the current hour. This is the same kind of baseline we used for demand.
- Ridge regression: a linear model with L2 regularization, which shrinks the coefficients to reduce overfitting when input features are correlated [5].
- Random forest: an ensemble of decision trees trained on random subsets of the data and features [6].
- Gradient boosting: builds decision trees one at a time, where each new tree tries to correct the errors of the trees before it [7]. This was the best model in our demand study.

For this renewable study, we used the same four non-LSTM models so that the comparison remained focused and reproducible. We did not include the LSTM because it did not beat the baseline in our demand study and we did not have time to tune it properly for a second target.

3. Methodology

3.1 Data Collection

We used the U.S. Energy Information Administration API to get hourly solar and wind generation data for CAISO in 2025 [1]. We used the same data source and basic process as in our demand project, but this time we selected solar (SUN) and wind (WND). We added solar and wind together to create one renewable-generation value for each hour.

We also used historical weather data from Open-Meteo for the Sacramento area, including temperature, humidity, dewpoint, and wind speed [3].

The two series were added together into a single column, renewable_MW = solar_MW + wind_MW, and merged onto the same hourly timestamps as the demand data. Weather variables (temperature, humidity, dewpoint, and wind speed) came from the Open-Meteo Historical Weather API for a Sacramento-area location [3], exactly as in the demand study.

3.2 Building the Renewable Dataset

We started with the 8,544-hour dataset from our demand project. Some rows could not be used because the renewable forecast needed information from before and after each hour.

We removed the first 168 hours because our longest history went back seven days. We removed the last 24 hours because we needed to know the renewable output 24 hours later. This left 8,352 usable rows for the renewable forecast.

- 168 rows removed from the start. The longest lag feature, renewable_lag_168h, needs 168 hours (7 days) of prior renewable output before it is defined.
- 24 rows removed from the end. The prediction target, renewable output 24 hours ahead, needs 24 hours of future data, which does not exist for the last 24 hours of the dataset.

That gives 8,544 − 168 − 24 = 8,352 usable rows. This is slightly different from the demand dataset, which lost 48 rows at the end instead of 24 because a 48-hour target column was also created during preprocessing. The renewable pipeline only builds a 24-hour target, so only 24 rows are lost.

3.3 Feature Engineering

We tried to use information similar to what we used for demand so that the comparison would be fair. The model inputs included renewable output from earlier hours, the current renewable output, averages from recent hours, the hour of day, day of week, month, and the available weather information.

For the backtest, we used the weather observed at the time of each forecast. In a real forecasting system, this would need to be replaced with a weather forecast for the next day.

- Lag features: renewable_lag_24h, renewable_lag_48h, and renewable_lag_168h.
- Rolling statistics: a 24-hour rolling mean of renewable output.
- Current output: renewable_MW at the hour the forecast is made.
- Time features: hour of day, day of week, and month, encoded the same way as in the demand model.
- Weather features: temperature, humidity, dewpoint, and wind speed at the hour the forecast is made.

As in the demand study, weather features are taken at the same timestamp as the row they belong to. They are never shifted forward, so no weather observation from the future forecast window enters the model. This does mean the model assumes weather observed at the time of the forecast is available, which is true for a backtest on historical data but would require an actual weather forecast in a live system. We discuss this in Section 5.4.

We also made sure that the models could not predict negative renewable generation. This changed only 4 of the 1,441 test predictions.

3.4 Data Split and Evaluation

We trained the models on the earlier part of the data and tested them on the later part. This is important because, in a real forecast, we would know the past but not the future. The training set had 6,911 hours and the test set had 1,441 hours, covering October 30 through December 29, 2025. This was the same test period used in our demand project, so the two forecasts could be compared on the same hours.

We measured the models using MAE, RMSE, and MAPE. For renewable generation, we calculated MAPE only for the 1,433 test hours where actual output was above 200 MW. Overnight renewable output can be close to zero, which would make percentage errors misleading.

We evaluated all models with three metrics: mean absolute error (MAE) in MW, root mean squared error (RMSE) in MW, and mean absolute percentage error (MAPE). For renewables, MAPE was computed only on the 1,433 of 1,441 test hours where actual output was above 200 MW. Overnight hours have output near zero, and dividing by a number near zero produces percentage errors that are enormous and meaningless, so including them would have made MAPE impossible to interpret.

3.5 Model Settings

We used the same hyperparameters as the demand models, without retuning them for renewable data. This kept the comparison fair: persistence used a 24-hour lag; Ridge used alpha = 10.0; Random Forest used 300 trees with maximum depth 14; and Gradient Boosting used 300 trees, maximum depth 3, and learning rate 0.05. The complete notebooks are available in the project repository [11].

4. Results

4.1 Demand Forecasting Results

These are the results from our first study, repeated here for comparison. All figures are on the same 60-day test set used below.

Table 1. Demand forecasting results on the 60-day test set (from our first study).

Model	MAE (MW)	RMSE (MW)	MAPE (%)	vs. Baseline
Persistence baseline	1,049	1,451	4.28%	—
Ridge Regression	1,698	2,030	6.82%	−61.8% (worse)
Random Forest	727	1,024	2.96%	+30.8%
Gradient Boosting	**717**	**980**	**2.91%**	**+31.7%**
LSTM	1,064	1,418	4.37%	−1.4%

Gradient boosting was clearly best here, and the two tree-based models both beat the baseline by around 31%.

4.2 Renewable Forecasting Results

We then ran the same pipeline with renewable output as the target. Figure 1 summarizes the renewable forecasting performance across the four models.

*Table 2. Renewable generation forecasting results on the same 60-day test set. *MAPE computed only on the 1,433 of 1,441 hours with actual output above 200 MW.*

Model	MAE (MW)	RMSE (MW)	MAPE* (%)	vs. Baseline
Persistence baseline	1,153.8	1,784.7	45.88%	—
Ridge Regression	1,098.4	1,691.7	53.16%	+4.8% (best)
Random Forest	1,295.3	1,886.0	80.04%	−12.3% (worse)
Gradient Boosting	1,282.0	1,990.5	70.10%	−11.1% (worse)

Only ridge regression beat the persistence baseline, and only by 4.8%. Both tree-based models — including gradient boosting, which won the demand comparison by a wide margin — performed worse than simply repeating the value from 24 hours earlier. Random forest was 12.3% worse and gradient boosting was 11.1% worse.

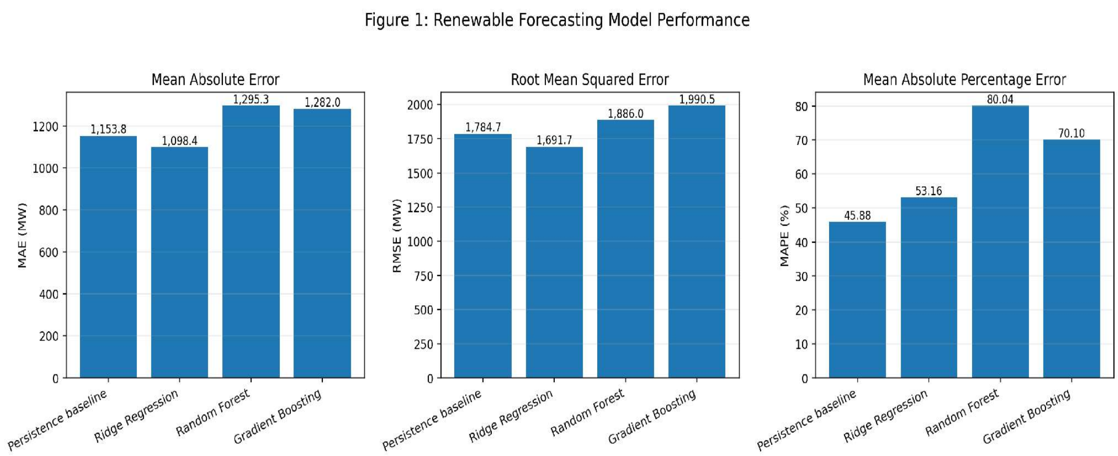

Figure 1. Renewable forecasting model performance for the 60-day test set.

We want to be direct about one thing here. An earlier version of our analysis reported the gradient boosting renewable error at about 1,283 MW without checking it against a baseline. That was a mistake on our part. Once we computed the persistence baseline, the number it should have been compared against was 1,153.8 MW, and gradient boosting loses to it.

4.3 Comparing the Two Problems

Putting the two studies side by side shows how different they are.

Table 3. Side-by-side comparison of the two forecasting problems.

	Demand forecasting	Renewable forecasting
Best model	Gradient Boosting	Ridge Regression
Improvement over baseline	+31.7%	+4.8%
Did tree ensembles beat baseline?	Yes, by ~31%	No, ~11–12% worse
Baseline error vs. mean output	4.1%	24.0%
Best-model error vs. mean output	2.8%	22.9%

The last two rows are the ones we find most telling. Mean demand over the year was about 25,621 MW and mean renewable output was about 4,802 MW, so we can express each error as a share of typical output. Our best demand forecast is off by about 2.8% of typical demand. Our best renewable forecast is off by about 22.9% of typical renewable output — roughly eight times worse in relative terms. Renewable generation is simply a much harder thing to predict a day ahead.

4.4 What the Gradient Boosting Model Was Using

To understand why gradient boosting struggled, we looked at which features it relied on. Figure 2 visualizes the same feature-importance results.

Table 4. Feature importance for the gradient boosting renewable model.

Feature	Importance
renewable_MW (current hour)	66.8%
renewable_lag_48h	29.5%
renewable_lag_168h	0.9%
temperature_C	0.5%
all other features (12 total)	2.3%

Current renewable output and the value from 48 hours earlier account for 96.3% of the model's feature importance together. That is worth pausing on: the model spent almost all of its capacity on two features that are essentially persistence-style information. It was building a very complicated path to a simple answer.

The weather features barely register. Temperature accounts for 0.5% of feature importance, while the remaining weather and time features are contained within the 2.3% grouped total. We think the relatively small weather contribution may be because none of the available variables is a good stand-in for cloud cover, which is the variable that actually drives solar output. Our dataset does not include a cloud-cover field.

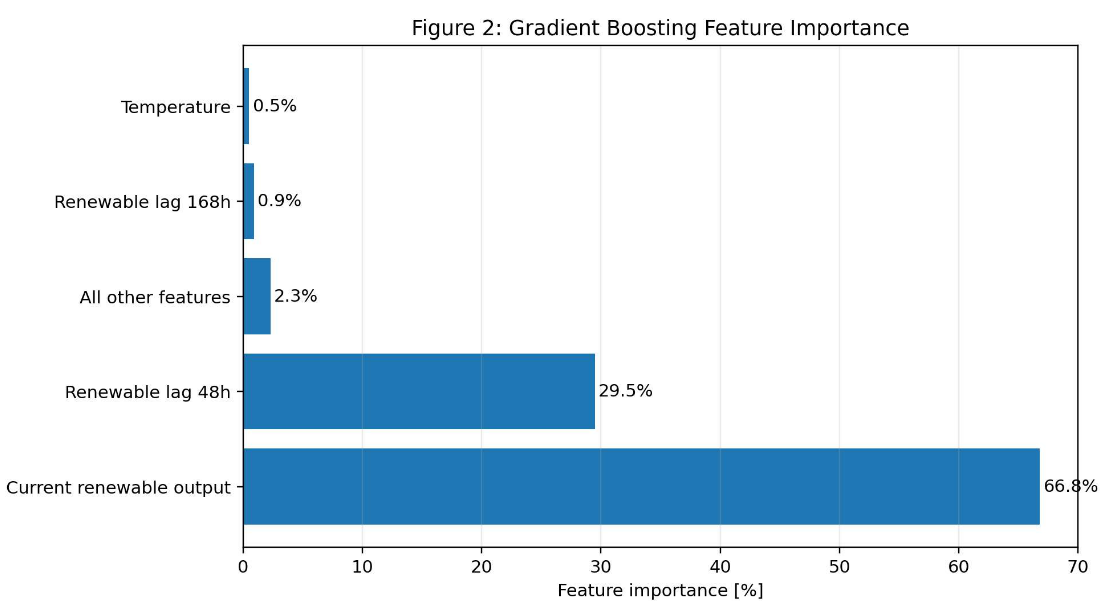

Figure 2. Gradient Boosting feature importance for the renewable forecasting model.

4.5 A Note on MAPE

Ridge regression has the best MAE and the best RMSE, but its MAPE (53.16%) is worse than the baseline's (45.88%). These two rankings disagree, and we do not want to hide that.

We think this happens because MAE and RMSE are driven mostly by the busy midday hours, when renewable output is high and the errors in megawatts are biggest. MAPE treats every hour above 200 MW the same, including early-morning and late-evening hours where a small miss in megawatts is a large miss in percent. So Ridge seems to do better on the big hours and worse on the small ones. We think the size of the error in megawatts matters more to a grid operator than the percentage, so we used MAE as our main measure. That is our choice, though, and someone else could reasonably decide the other way.

5. Discussion

5.1 Why Renewable Generation May Be Harder to Forecast

Our results suggest demand and renewable generation are not just the same kind of data at different sizes. They behave differently.

Demand is driven by human routine, and routines repeat. The same hour on the same weekday tends to look similar week to week, which is exactly the sort of structure a model can learn. Weather shifts the level up or down, but the shape stays.

Renewable generation depends on the weather, and the weather does not repeat on a schedule. Solar does have one strong pattern that a model can learn, the daily arc from sunrise to sunset. But once you take that pattern out, most of what is left comes from clouds. Wind has almost no time-of-day pattern to learn in the first place. When we add solar and wind together into one number, a lot of the variation is simply very hard to predict a day ahead.

5.2 Why the More Complex Models Did Worse

We cannot prove exactly why gradient boosting and random forest underperformed, since our experiment measures how models performed rather than why. But we can offer a possible explanation that fits what we observed.

If most of the useful information in renewable output is already close to persistence, then the best available answer is a simple one. Ridge Regression is a straight-line model with regularization, so it can give that simple answer directly. Gradient Boosting has to get to the same place by building 300 decision trees (maximum depth 3, learning rate 0.05), and each tree can latch onto patterns that showed up in the training period but do not repeat in the test period. When the data is this noisy, that flexibility hurts more than it helps.

The feature importance table in Section 4.4 is consistent with this. Gradient boosting assigned 96.3% of its feature importance to current output and the 48-hour lag, which suggests it was trying to approximate persistence-like behaviour through a much more flexible and more overfittable path. Ridge got to a similar answer more directly, and did slightly better.

If this explanation is right, it is not really about gradient boosting being a bad algorithm. It is about matching how complicated a model is to how much of a pattern the data actually has.

5.3 Combining the Two Forecasts

Even an imperfect renewable forecast is useful, because what grid operators actually plan around is net load, which is demand minus renewable generation. Using the Gradient Boosting demand forecast together with the Ridge renewable forecast, we can estimate net load 24 hours ahead instead of just demand. Figure 3 shows when the test period produced high-stress and renewable-surplus signals.

Our demand model is off by about 717 MW on average and our renewable model by about 1,098 MW. If those two errors were completely independent, the combined net-load error would land somewhere between the larger of the two and their sum, depending on whether the misses line up. We have not actually measured how related the two error series are, so we are not going to give a combined number we did not test. Measuring that is one of the first things we want to do next. Figure 4 shows a one-week sample of the predicted demand and renewable series.

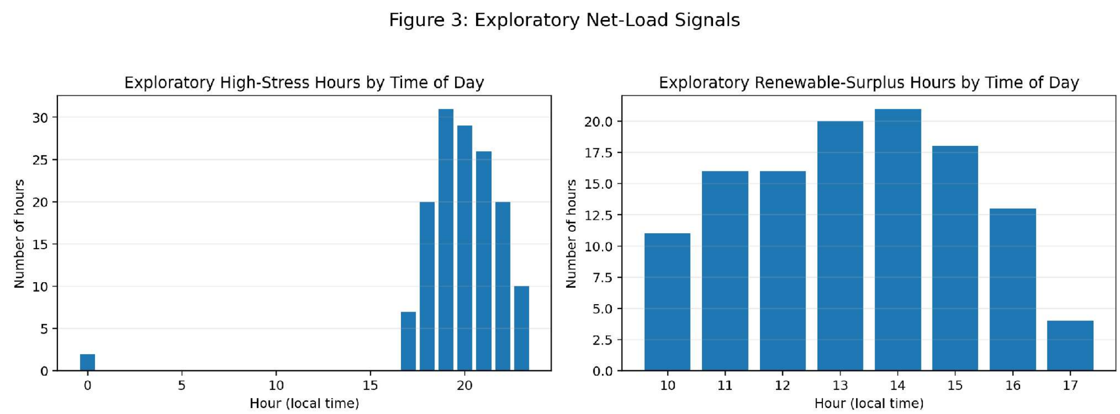

Figure 3. Exploratory high-stress and renewable-surplus hours by local time across the test period, using Gradient Boosting demand predictions and Ridge renewable predictions.

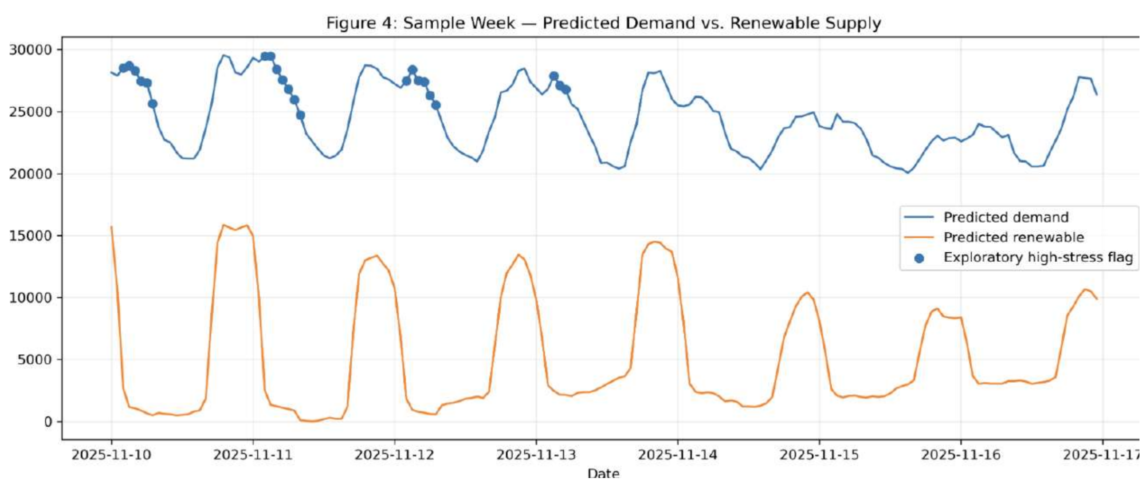

Figure 4. One-week sample of Gradient Boosting predicted demand and Ridge predicted renewable supply; exploratory backtest visualization.

What we can say is that the renewable forecast is now the weaker link. Improving it would do more for a net-load forecast than further improving the demand model, and that is a useful thing to know before starting the next project. The exploratory plots in Figures 3 and 4 are not operational dispatch results.

5.4 Limitations

- One region and one test window. All results come from CAISO in 2025, tested on October 30 through December 29. That window is autumn and early winter, when solar output is lower and less variable than in summer. Results could look different on a summer test fold, and we have not run one.
- Solar and wind were combined into one target. They have very different behaviour — solar is highly structured by time of day, wind is not — and averaging them into a single number may make both harder to predict. Training separate models is an obvious next step.
- No cloud-cover feature. Our weather data does not include cloud cover, which is probably the single most useful variable for solar forecasting. Adding it would likely help more than any weather feature we currently have.
- Observed weather, not forecast weather. The models use weather measured at the time the forecast is made. A live system would have to use a weather forecast for the target hour, which carries its own error. Our numbers should be read as an optimistic case.
- We did not compare against professional utility forecasting systems. Our comparison is against a naive persistence baseline and the specific models listed in this paper, and nothing more.
- Hyperparameters were not tuned for renewable data. This was deliberate for fairness, but it means a properly tuned gradient boosting model might do better than the one we report.

One more thing worth saying: we tried to give both models similar inputs, but they were not identical, because demand and renewable generation are different kinds of data. So we cannot say for certain that the gap in results comes only from renewables being harder to predict. Testing both with exactly the same inputs would be a good next step.

6. Conclusion and Future Work

We asked whether the machine-learning models that improved our 24-hour electricity demand forecast would also improve a 24-hour renewable generation forecast. Using 2025 CAISO solar and wind data, the same feature pipeline, the same 60-day test window, and the same four-model settings, we compared a persistence baseline, ridge regression, random forest, and gradient boosting.

They did not transfer. For demand, gradient boosting reduced mean absolute error by 31.7% relative to the naive baseline. For renewable generation, random forest was 12.3% worse and gradient boosting was 11.1% worse than the baseline, while only ridge regression beat it, by 4.8%. Among the models we tested, the simplest one won on the harder problem and lost on the easier one.

Our results suggest that how much of a pattern the data actually has should guide how complicated a model to use, rather than reusing whichever model worked on a previous problem. They also suggest that predicting renewable generation a day ahead, at least the way we tried it, is a much harder problem than predicting demand. Our best renewable forecast was off by about 23% of typical renewable output, compared with about 3% for demand.

For future work, we want to test solar and wind as separate targets, add a cloud-cover forecast feature, and evaluate on a summer test fold where renewable output is larger and more variable. We would also like to measure how the demand and renewable forecast errors interact when they are combined into a net-load forecast.

Beyond that, we want to build toward a dispatch question. Small Modular Reactors (SMRs) are being discussed as flexible clean-power resources [9]. A day-ahead net-load forecast could, in principle, help identify when flexible generation is most needed. We have not modelled SMR operations, dispatch, capacity factors, costs, or emissions; those outcomes are outside this study. Building and testing a dispatch optimization on top of these forecasts is a next step.

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